FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Advanced)-2021

PAPER –2

TEST DATE: 23-12-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. AC

2. BC

Sol. $$E_{inside}\,4\pi r^2 = \frac{\int_0^x \left(\frac{5}{2\pi}\right)\left(1-\frac{r}{30}\right)4\pi r^2 dr}{\varepsilon_0}$$

$$\frac{dE_{inside}}{dx} = 0$$

3. C

Sol. $$v_e = \sqrt{2}\sqrt{\frac{GM}{r_0}}$$

$$\therefore \Delta v = v_e - v_0 = \left(\sqrt{2}-1\right)\sqrt{\frac{GM}{r_0}}$$

4. ABC

Sol. Positive power lens (convex lens in rare medium) can from enlarged (or of same size) real, inverted image.
Positive power lens (convex lens in rare medium) can form virtual, erect, diminished image.

5. BD

Sol. $-\mu mgx = \frac{1}{2}mv^2 - \frac{1}{2}mv^2$

$\Rightarrow -\frac{1}{8}(20)(1) = \frac{1}{2}(2)v^2 - \frac{1}{2}(2)5^2$

$\Rightarrow v = \frac{3}{2}\sqrt{5} m/s$

$-\mu mgx = 0 - \frac{1}{2}mv^2$

$x = 10m$

Total distance $= 20m$

6. ACD

Sol. $-\frac{1}{4}\times 20\times x - \left(\frac{1}{2}50x^2\right) = 0 - \frac{1}{2}2(6/5)^2$

$\Rightarrow x = \frac{8}{25}m = 32cm$

$a = \frac{kx - f_s}{m} = \frac{16-5}{2} = 5.5 m/s^2$

$\& \; -\frac{1}{4}(20)(8/25) - \left(-\frac{1}{2}50\times\frac{64}{625}\right) = \frac{1}{2}(2)v^2$

$\Rightarrow v^2 = 24/25 \Rightarrow U = \frac{2\sqrt{6}}{5} m/s$

SECTION – B

7. 2

Sol. $R = 50\Omega$

$X_L = \omega L = 50\Omega$

$X_L = \frac{1}{\omega C} = 100\Omega$

$\Rightarrow Z = \sqrt{R^2 + (X_L - X_C)^2}$

$Z = 50\sqrt{2}\Omega$

$\mathrm{I_{ram}} = \frac{V_{rms}}{Z} = \frac{200/\sqrt{2}}{50\sqrt{2}} = 2A$

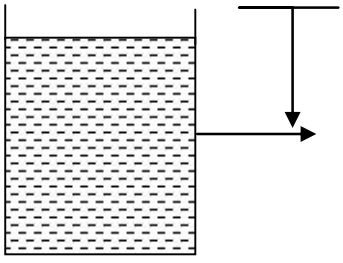
8. 5

Sol. $V = \sqrt{2gy}$

$\Rightarrow dF = \left(1\times 10^{-3}\right)\left(\sqrt{2gy}\; dy\sqrt{2gy}\times\rho\right)$

$dF = 1\times 10^{-3}\times 2\times 10\times y\times 10^3\; dy = 40\; ydy$

$\Rightarrow F = \frac{20}{2}\int ydy = \frac{40}{2}\left(0.75^2 - 0.25^2\right) = 20\times 0.5 = 5$



9. 9

10. 1

Sol. $\frac{dx}{dt} = a \Rightarrow \frac{d^2x}{dt^2} = 0$

$\frac{dy}{dt} = a - 2a\alpha t \Rightarrow \frac{d^2y}{dt^2} = -2a\alpha$

Velocity vector $\vec{v} = a\hat{i} + (a - 2a\alpha t)\hat{j}$

$\vec{a} = -2a\alpha\hat{j}$

$\vec{v}.\vec{a} = va\cos\frac{\pi}{4}$

$t = \frac{1}{\alpha} \Rightarrow A = 1$

11. 0

Sol. Block start when $\mu_s mg = 4t$

$t = 9\,\text{sec}$

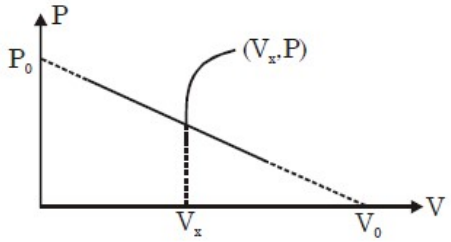
At $t = 5s$ block was at rest so answer = 0

12. 4

SECTION – C

13. 00000.58

Sol. $\frac{P}{V_0 - V_x} = \frac{7}{5}\frac{P}{V_X}$



$5V_x = 7V_0 - 7V_x$

$V_x = \frac{7V_0}{12}$

14. 00400.00

Sol. IIIrd overtone

$f = \frac{4V}{2L} = \frac{2V}{L} = \frac{2}{1}\sqrt{\frac{200}{(5/1000)}} = 2(200)$

$f = 400Hz$

15. 00014.00

Sol. $u=-\left(4+\frac{4}{2}\right)\times 2=-12\,cm$

$\frac{1}{v}+\frac{1}{-12}=\frac{1}{-8} \quad \Rightarrow v=-24\,cm$

For final image (refraction through plane surface) we can write.

$24=2\left(\frac{x}{1}+\frac{4}{2}\right)\Rightarrow \quad 24=2x+4$

$x=10\,cm$

So distance from P is $14\,\text{cm}$

16. 00002.00

Sol. Let width of second strip is ℓ

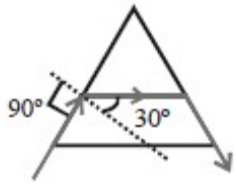
$\left(S_1O-4\right)+1.5\left(4\right)=\left(S_2O\text{-}\ell\right)+\left(\ell+\frac{\ell^2}{2}\right)$

$\left(1.5-1\right)\left(4\right)=\frac{\ell^2}{2}$

$\ell=2mm$

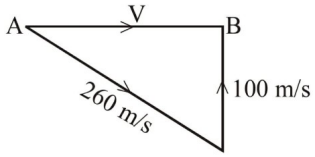
17. 00002.00

Sol. $\mu=\frac{\sin 90^\circ}{\sin 30^\circ}=2$



18. 00001.25

Sol. Speed of sound in direction of AB $V=\sqrt{260^2-100^2}=240\,m/s$



$\therefore t=\frac{300}{240}=1.25$

Chemistry

PART – II

SECTION – A

19. CD

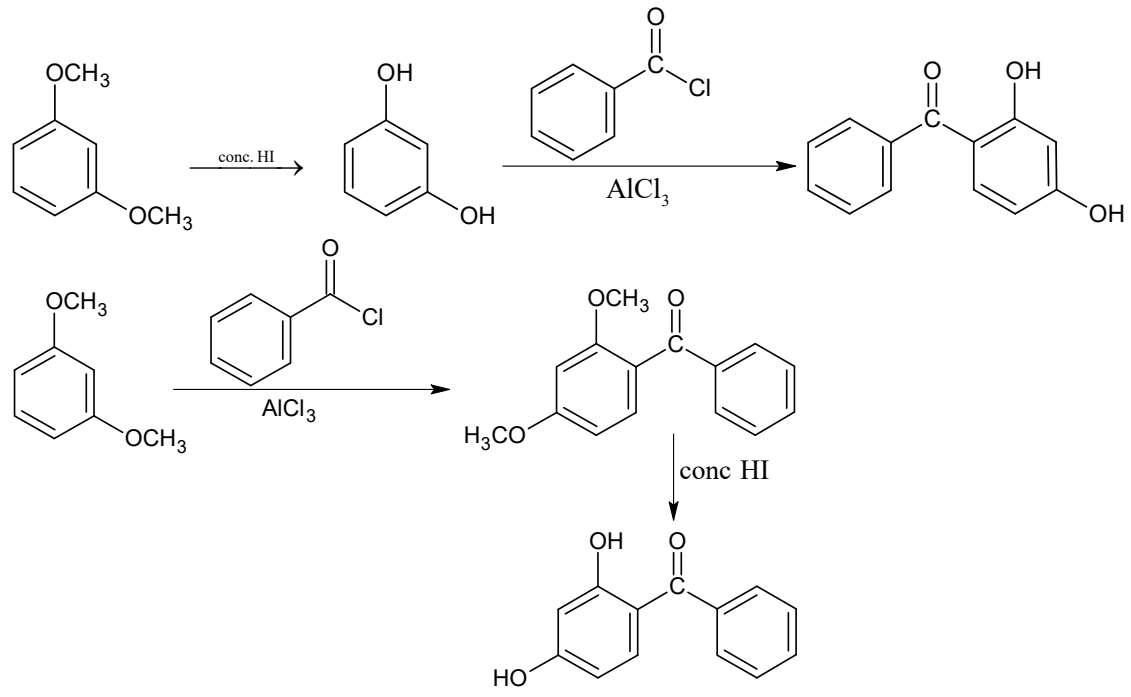
20. AB
Sol. Solute X will dissociate into the solution so i>1.

21. ABD
Sol. $\left[CoF_6\right]^{3-}$ is a high spin complex.

22. ABCD

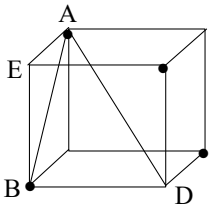
23. ACD
Sol. Due to the more cohesive energy in Li, it has highest melting point among alkali metals.

24. BC
Sol.



SECTION – B

25. 6
Sol.



Face diagonal of cube $(AB)=2R$

Edge length of cube $(BE)=\frac{2R}{\sqrt{2}}$

Body diagonal of cube $(AD)=\sqrt{3}\times\frac{2R}{\sqrt{2}}$

$=\sqrt{6}\times R$

26. 4

Sol. $pKa\ of\ HX=14-10=4$

$pH=pK_a+\log\frac{1}{1}=4$

27. 8

Sol. $(CO)_{x/2}\,Co-Co(CO)_{x/2}$

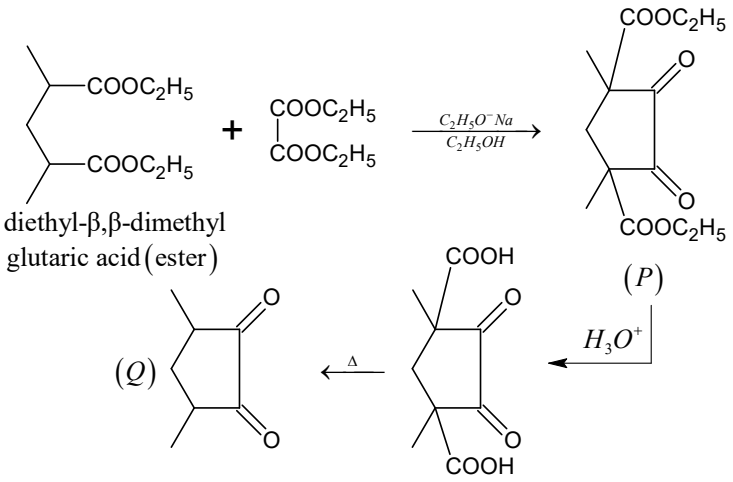
$EAN(=36)=27+1+2\times\frac{x}{2}$

$\Rightarrow 28+x=36$

$x=8$

28. 6

Sol.



29. 4

Sol. $\frac{1}{2}H_2(g)+\frac{1}{2}Cl_2(g)\rightleftharpoons HCl(g)\,\Delta G^\circ=1.72\,kJ\,mol^{-1}$

ΔG° for $2HCl(g)\rightleftharpoons H_2(g)+Cl_2(g)$ is $\left[2\times(-1.72)\right]kJ$

$\Delta G^\circ=-2.303RT\log K_p$

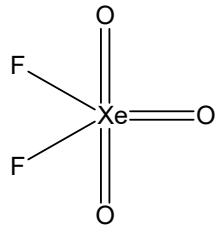
$-3.44\times10^3=-5700\log K_p$

$\log K_p=0.6$

$K_p=4$

30. 6

Sol.



SECTION – C

31. 00500.00

32. 00100.00

Sol. $K = Ae^{-Ea/RT}$

33. 09900.00

Sol. $\Delta U = q\ tw$

For adiabatic process $q = 0$

$\Delta U = w$

$\Delta U = P\Delta V = -P\left(V_2 - V_1\right)$

$\Delta U = -100\left(99 - 100\right) = 100\,\text{bar mL}$

Now $\Delta H = \Delta U + \Delta\left(PV\right)$

$\Delta H = 100 + \left[100 \times 99 - 1 \times 100\right]$

$\Delta H = 9900\,\text{bar mL}$

34. 00008.00

35. 00006.00

36. 01501.00

Mathematics

PART – III

SECTION – A

37. BD

38. D

39. BCD

Sol. $p(n)$ = Probability that child stops after drawing exactly n marbles

$$p(n)=\frac{1}{2}.\frac{2}{3}.\frac{3}{4}.....\left(\frac{n-1}{n}\right)\left(\frac{1}{n+1}\right)=\frac{1}{n(n+1)}$$

40. ABCD

Sol. $x^3+2x^2-3x+1=0\begin{cases}\alpha\\ \beta\\ \gamma\end{cases}$

$$x^3-3x^2-2x+1=0\begin{cases}1/\alpha\\ 1/\beta\\ 1/\gamma\end{cases}$$

Given expression $\frac{1}{3-\frac{1}{\alpha}}+\frac{1}{3-\frac{1}{\beta}}+\frac{1}{3-\frac{1}{\gamma}}$

$$=\frac{\sum\left(3-\frac{1}{\alpha}\right)\left(3-\frac{1}{\beta}\right)}{\left(3-\frac{1}{\alpha}\right)\left(3-\frac{1}{\beta}\right)\left(3-\frac{1}{\gamma}\right)}=\frac{27-6\sum\left[\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}\right]+\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}}{\left(3-\frac{1}{\alpha}\right)\left(3-\frac{1}{\beta}\right)\left(3-\frac{1}{\gamma}\right)}=\frac{11}{7}$$

41. ABC

Sol. $3x^2\frac{dx}{dy}=\frac{x^3}{x^3-y}\qquad x^3=t$

$$\Rightarrow\frac{dt}{dy}=\frac{t}{t-y}\quad\Rightarrow(t-y)dt=t\,dy$$

$$\Rightarrow t\,dt=y\,dt+t\,dt\Rightarrow\frac{t^2}{2}=t.y+\frac{c}{2}$$

$$\Rightarrow x^6=2x^3y+c$$

42. AD

Sol. Let

x: Total No of ways of selecting even number of red balls

even number of red balls

y: Total No of ways of selecting

odd numbers of red balls

$(2-1)(4-1)(6-1)(8-1)(10-1)=x-y \quad (2+1)(4+1)(6+1)(8+1)(10+1)=x+y$

SECTION – B

43. 9

Sol. A.M.=G.M.

$$\frac{r_1}{2}=\frac{r_2}{4}=\frac{r_3}{5}=\frac{r_4}{8}=k$$

44. 8

Sol. $t_1+t_2=2$

$$\frac{2-m_2}{1+2m_2}=\sqrt{3}$$

$$\text{and}\ \frac{m_3-2}{1+2m_3}=\sqrt{3}$$

$$t_1+t_2+t_3=\frac{p}{q}=\frac{3}{11}$$

45. 2

46. 0

47. 6

48. 3

SECTION – C

49. 00004.00

50. 00008.00

Sol. The triangle has circumcentre at origin and its orthocentre laying on the circumcircle.

51. 00005.00

Sol. $$\lim_{y\to 1}\frac{100y^{100}}{y^{100}-1}-\frac{50.y^{50}}{y^{50}-1}$$

$$\lim_{y\to 1}100-50-\left(\frac{100}{1-y^{100}}-\frac{50}{1-y^{50}}\right)$$

$$L=50-L\ \therefore L=\frac{50}{2}=25$$

52. 00001.00

Sol. $$T_n=\frac{H_{n+1}-H_n}{H_nH_{n+1}}$$

As $H_{n+1}-H_n=\frac{1}{n+1}$

$\Rightarrow T_n=\frac{1}{H_n}-\frac{1}{H_{n+1}}$

$\Rightarrow T_1=\frac{1}{H_1}-\frac{1}{H_2}$

$T_2=\frac{1}{H_2}-\frac{1}{H_3}$

$\vdots \quad \vdots \quad \vdots$

$T_n=\frac{1}{H_n}-\frac{1}{H_{n+1}}$

$T_1+T_2+...+T_n=\frac{1}{H_1}-\frac{1}{H_{n+1}}$

$\text{As } n\to\infty, \quad T_1+T_2+...+0=1$

53. 00007.00

Sol. Clearly a team of 4 with two batsman and two bowler can be formed as

Case-I: 2 bowler+2batsman

Number of ways= ${}^4C_2\times{}^4C_2=36$

Case-II: 2 bowler+1 batsman +1 all-rounder

${}^4C_2\times{}^4C_1\times{}^1C_1=24$

Case-III: 1bowler +2 batsman +1 all-rounder

${}^4C_1\times{}^4C_2\times{}^1C_1=24$

$\therefore$ Probability that all-rounder is selected= $\frac{24+24}{36+24+24}=\frac{4}{7}$

$\therefore \quad p=7$

54. 00005.00